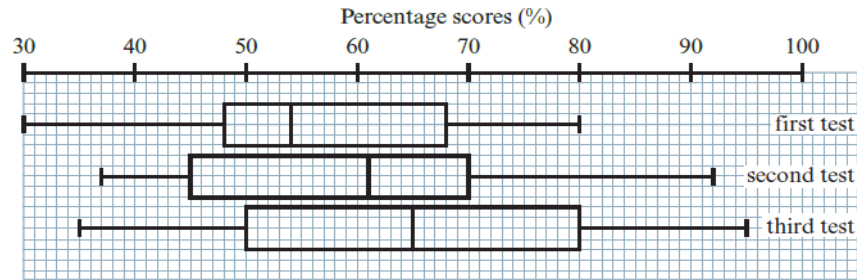


- Q1** (a) The following box plots summarize the percentage scores of a class of students in the three Mathematics tests they took this term



- (i) Find the median and interquartile range of the first test score. [2]

$$Me = 54, IQR = 68 - 48 = 20$$

- (ii) How much higher is the upper (third) quartile of the third test compared to the first test? [1]

$$Q_3^3 - Q_3^1 = 80 - 68 = 12$$

- (iii) Showing necessary working state, the nature of the distribution of the second test scores? [2]

$$Q_1 \sim Me = 45 \sim 61 = 16 > 9 = 61 \sim 70 = Me \sim Q_3 \rightarrow \text{Left-skewed}$$

- (b) The first four central moments of distribution are 0, 7, 26 and 39 respectively. Calculate the standard deviation and skewness and hence comment on the nature of the distribution. [2]

$$\sigma = \sqrt{m_2} = \sqrt{7}$$

$$\gamma_3 = \frac{m_3}{\sqrt{m_2^3}} = \frac{26}{\sqrt{7^3}} = 1.4 > 0 \rightarrow \text{Right-skewed}$$

- (c) Survey of a random sample of 20 people recorded the number of unwanted emails they received in a particular week. The results are given below.

18 13 18 21 17 22 27 8 11 26
32 17 31 20 36 15 13 25 35 29

Represent the data in a stem and leaf diagram and hence find the median. [3]

Stem	Leaf
0	8
1	1, 3, 3, 5, 7, 7, 8, 8
2	0, 1, 2, 5, 6, 7, 9
3	1, 2, 5, 6

Key: 3|1 → 31

$$Me = \frac{20 + 21}{2} = 20.5$$

Q2

(a) Using the data given in the above table

Height h (cm)	$0 \leq h < 10$	$10 \leq h < 20$	$20 \leq h < 30$	$30 \leq h < 40$	$40 \leq h \leq 50$
Frequency	3	12	9	6	5

(i) Identify the median class and find the percentage whose heights are below 20 cm. [2]

Class	f_i	F_i
$0 \leq h < 10$	3	3
$10 \leq h < 20$	12 (Modal)	15
$20 \leq h < 30$	9	24 (Median)
$30 \leq h < 40$	6	30
$40 \leq h \leq 50$	5	35

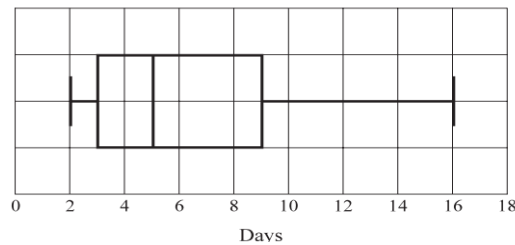
$$\frac{N}{2} = \frac{35}{2} = 17.5 \rightarrow 20 \leq h < 30$$

$$\% \rightarrow \frac{3 + 12}{35} \times 100\% = 42.86\%$$

(ii) Calculate the mode of the distribution. [3]

$$Mo = 10 + \frac{(12 - 3)}{(12 - 3) + (12 - 9)} \times 10 = 17.5$$

(b) A health worker plotted a box-and whisker plot of the times that patients stayed in hospital.



Based on the diagram, determine the possible range of times that could be considered as outliers. [2]

$$\text{Outliers for } t < 2 \text{ or } t > 16. \text{ So, required range } R = 16 - 2 = 14$$

(c) Two shopkeepers sell rice.

Shopkeeper A records the daily sales (in kg) over 5 days as: 48, 52, 50, 49, 51.

Shopkeeper B records the mean daily sales over the 5 days of is 50 kg and standard deviation is 10 kg.

Calculate the **Coefficient of Variation (CV)** for both shopkeepers and determine whose sales are more consistent. [5]

Here, $\sum x_A = 250$ and $\sum x_A^2 = 12510$

$$\bar{x}_A = \frac{\sum x_A}{5} = \frac{250}{5} = 50$$

$$\sigma_A = \sqrt{\frac{\sum x_A^2}{5} - \left(\frac{\sum x_A}{5}\right)^2} = \sqrt{\frac{12510}{5} - \left(\frac{250}{5}\right)^2} = 1.41$$

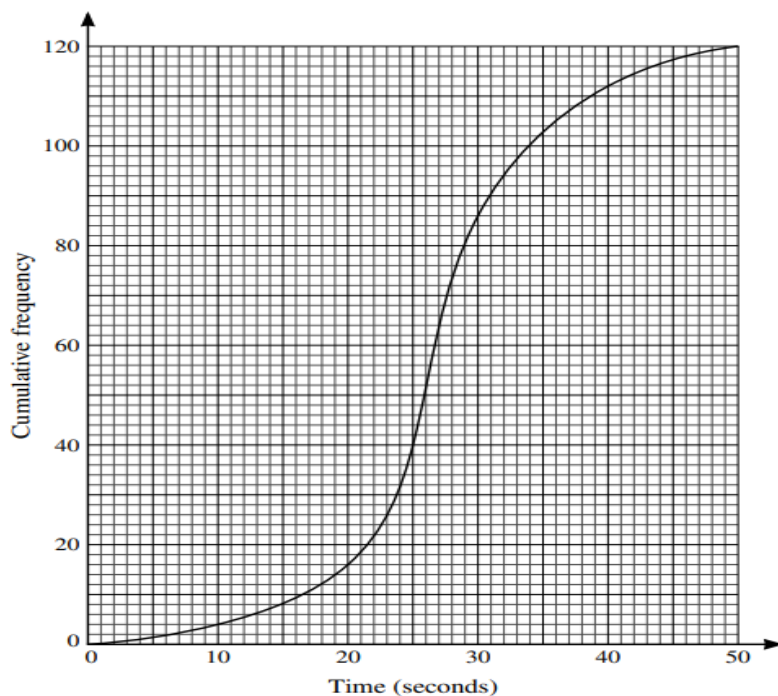
$$CV_A = \frac{\sigma_A}{\bar{x}_A} \times 100\% = \frac{1.41}{50} \times 100\% = 2.83\%$$

$$\bar{x}_B = 50 \text{ and } \sigma_B = 10$$

$$CV_B = \frac{\sigma_B}{\bar{x}_B} \times 100\% = \frac{10}{50} \times 100\% = 20\%$$

$CV_A < CV_B \rightarrow A$ is more consistent.

- Q3** (a) The times taken by 120 children to complete a particular puzzle are represented in the following cumulative frequency graph.



- (i) Use the graph to estimate how many of the students took between 20 to 30 seconds. [1]

$$86 - 16 = 70$$

- (ii) 30% of the children took more than T seconds to complete the puzzle. Use the graph to estimate the value of T . [2]

$$P_{(100-30)} = P_{70}: \frac{70 \times 120}{100} = 84 \rightarrow P_{70} = 29.5$$

- (b) A company wants to analyze the relationship between the number of hours studied (x) and the score obtained in a test (y) by 6 students. The data collected is as follows:

Hour Studies (x)	2	3	5	8	9	10
Test Score (y)	65	70	75	85	95	98

- (i) Using the method of least squares, find the equation of the linear regression line in the form:
 $y = a + bx$ [4]

x	y	x^2	xy
2	65	4	130
3	70	9	210
5	75	25	375
8	85	64	680
9	95	81	855
10	98	100	980

$$\sum x = 37$$

$$\sum y = 488$$

$$\sum x^2 = 283$$

$$\sum xy = 3230$$

$$b_{\frac{y}{x}} = \frac{6 \times 3230 - 37 \times 488}{6 \times 283 - 37^2} = 4.024$$

$$\hat{y} = 4.024 \left(x - \frac{37}{6} \right) + \frac{488}{6} = 4.024x + 56.509$$

- (ii) Using the above equation, determine the number of study hours required to achieve a score of 90. [1]

$$90 = 4.024x + 56.509 \rightarrow x = \frac{90 - 56.509}{4.024} = 8.32$$